Ex: 1, 3, 7, 5 Error: $\frac{1}{2}(\cancel{4}+0+4+4)^2$ Error(3) $= \frac{1}{2}(3+1-3-1)^2 = 0$ Error: $\frac{1}{2}(4+2-2+0)^2$
mean is 4, $= \frac{1}{2}(-4)^2 = 8$ $= 8$
guess is 3 guess: 4 guess: 5

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#3

Supp #3,4,13

R#9-12

(3) Our gradient descent rule is that the error for this function needs to be 0. Once the error is 0, we have found the mean. By taking the derivative of the error function w/ respect to $\hat{x}$, we get $\sum_i (\hat{x}-x_i)$. If this is negative, our $\hat{mean}$ is below the true mean, if its positive, it's greater than it. So, using $\hat{x} = \hat{x} - p\,\partial Error(\hat{x})$ or $\hat{x} = \hat{x} - p(\sum_i (\hat{x}-x_i))$, ~~by~~ setting a small learning rate p we can achieve the mean

(4) $x, y = (1.3, 5.4)$, $f(x,y) = x^2 + xy + y^2$ SGD for x: GGD for y:

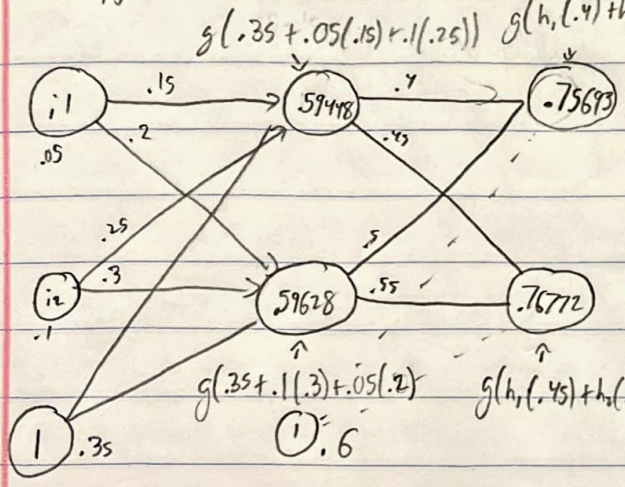
$\frac{\partial}{\partial x} f(x,y) = 2x+y$ $\hat{x} = 1.3 - .01(2\cdot 1.3 + 5.4)$ $\hat{y} = 5.4 - .01(1.3 + 2\cdot 5.4)$

$\frac{\partial}{\partial y} f(x,y) = x + 2y$ $\hat{x} = 1.22$ $\hat{y} = 5.279$

Descent: $\hat{x} = \hat{x} - \alpha\left(\frac{\partial}{\partial x} f(x,y)\right)$

(13) $\frac{\partial R_i(\theta)}{\partial W_{kj}} = -(y_i - f_\theta(x_i)) \cdot \beta_k \cdot g'(Z_{ik}) \cdot x_{ij}$ $g(x) = sigmoid(x) = \frac{1}{1+\frac{1}{e}}$

$g'(x) = \sigma(x)\cdot(1-\sigma(x))$



$O_1 = .7569319$ $Z_{1,2} = .6 + .4(h_1) + .5(h_2) = O_1$

$k=2$ $O_2 = .7677179$

$\beta_2 = \beta_2 - .5(-(y_1 - f_\theta(x_1)) \cdot g(Z_{1,2}) - (y_2 - f_\theta(x_2)) \cdot g(Z_{2,2})$

$= .6 - .5(-(.01 - O_1) \cdot g(O_1) - (.99 - O_2) \cdot g(O_2)$

$= .421699$

$k=2, j=1$

$W_5 = W_5 - .5(-(y_1 - f(x_1)) \cdot \beta_2 \cdot g'(O_1) \cdot h_1 - (y_2 - f(x_2)) \cdot \beta_2 \cdot g'(O_2) \cdot h_1)$

$= .385683$

$\theta^m = \langle w_1, w_2, w_3, w_4, b_1, \ldots, b_2 \rangle$ There's many more weights... this ball I can feasibly do.